

Predicting 30-Day Hospital Readmission

BA865 | Team 08

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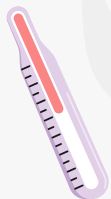
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01

Problem & Motivation

+ What is 30-day readmission?

DEFINITION

An unplanned hospital admission occurring within 30 days of a prior discharge.

Our Goal: Predict it before it happens.

Accurate risk stratification at discharge time gives clinicians a window to intervene before a patient deteriorates at home — the kind of decision support that turns reactive medicine into preventive medicine.

WHAT A GOOD PREDICTION ENABLES

Schedule

Early follow-up appointments before symptoms recur

Adjust

Discharge medications and reconcile prescriptions

Arrange

Home health visits and post-acute care coordination

Avoid

CMS reimbursement penalties



20%

Medicare patients readmitted within 30 days

\$17B

Annual U.S. readmission cost of avoidable admissions

3%

Max CMS Reimbursement penalty



02 MIMIC-IV Dataset



+ Dataset · MIMIC-IV v3.1

Beth Israel Deaconess Medical Center · 2008–2019 · de-identified

CONTENT

65,000+

ICU patients

200,000+

ED patients

Discharge

Summary notes

Labs & Vitals

ICD-10, meds, demographics

COHORT FILTER

Adults 18+

With discharge summary

≈50-80K

Admissions

≈40K

Unique patients

Excludes

In-hospital deaths and transfers

ACCESS

Physionet

Credential access

CITI Training

Human-subject research

DUA signed

Data use agreement

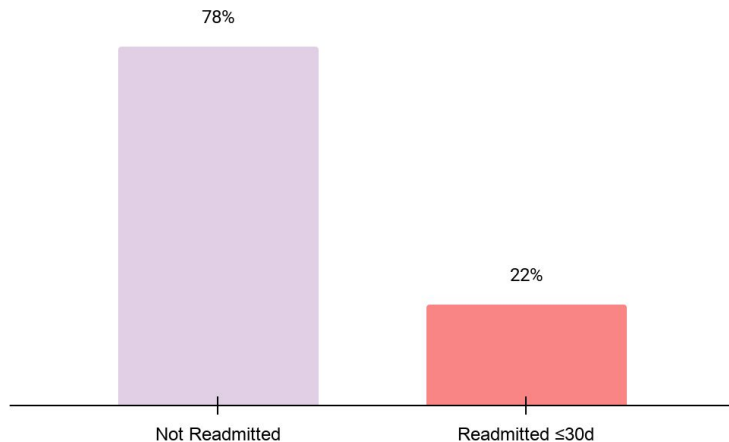
Patient-level

Splits to prevent leakage



+ Cohort & Class Imbalance

Class Distribution (Admissions)



Source: MIMIC-IV admissions table; positive class defined by next admission ≤30 days.

KEY INSIGHTS

4 : 1

Negative-to-positive ratio

Plain accuracy is misleading — a 'never readmit' baseline scores 80%.

Class_weight

{0:1, 1:3}

Triple the loss penalty on missed readmissions to counter imbalance.

AUC + Recall

Primary Metrics

AUC is threshold-free and robust; recall is the clinical priority.





03

Neural Network Architecture

+ Multimodal Inputs

CLINICAL TEXTS

Roberta - SBERT and ClinicalBERT

Discharge summaries: physician notes with diagnoses, medications, clinical reasoning, and follow-up instructions.

EXAMPLES

- Tokenized to 512
- Frozen backbone

NUMERICAL

via StandardScaler

Continuous features measured at admission and during the stay — vitals, labs, length of stay, prior admission count.

EXAMPLES

- Patient age
- Length of stay
- Creatine, hemoglobin

CATEGORICAL

via LabelEncoder

Discrete clinical and demographic codes that capture context the numerical features can't represent on their own.

EXAMPLES

- Gender, insurance
- Admission type
- ICD-10 chapter



Hyper-parameter exploration

Embedding Strategy

Biggest performance lever

- SBERT (384-dim)
- RoBERTa (768-dim)
- ClinicalBERT mean pool
- ClinicalBERT rich pool
- ClinicalBERT chunked

Architecture progression

Components added incrementally

- Class weights
- Dual-input (multimodal)
- BatchNorm + Dropout
- MultiHeadAttention
- Residual + dual pooling

Training settings

Tuned for stability and fit

- Dropout 0.3 → 0.4
- Attention heads 2 → 4
- LR 1e-3 + ReduceLROnPlateau
- F1-optimal threshold sweep
- Monitor val_auc → val_auc_pr

Held constant for fair comparison:

Adam optimizer

20% validation split

EarlyStopping patience 5

Class-weighted loss

Models tested

13

Encoders compared

3

Best Model

ClinicalBERT V2



V1. MultiModal RoBERTa Classifier

INPUTS

Discharge Notes (Text)



Text preprocessing pipeline:

- Tokenize with RoBERTa tokenizer
- Pad / truncate to 512 tokens
- Frozen RoBERTa backbone

Structured Features



Structured Processing:

- 10 numerical + categorical features
- StandardScaler on numerics
- LabelEncode categoricals

Feature Fusion



Concatenate the RoBERTa CLS embedding (256-dim) with the structured branch output (32-dim) inside the Keras Functional API — enables joint cross-modal learning

Classifier Head



Dense Classifier:

- Dense 64 + Relu
- Dropout 0.2 Regularization
- Dense 1 + Sigmoid

OUTPUTS

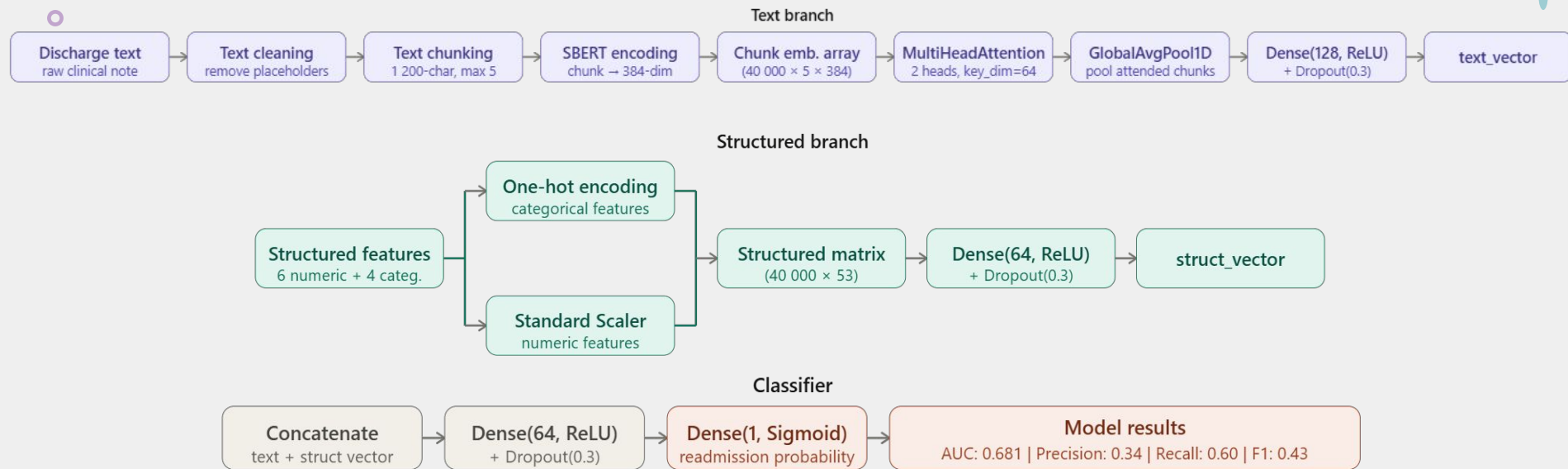
AUC: 0.69 | Precision: 0.37 | Recall: 0.52 | F1: 0.44





V2. MultiModal SBERT + Attention Classifier

Pipeline: Model's Architecture and Results



AUC: 0.681 | Precision: 0.34 | Recall: 0.60 | F1: 0.43

V3. MultiModal ClinicalBERT + Attention Classifier

How to improve our embeddings and model?



What is ClinicalBERT?

ClinicalBERT is a specialized BERT-based language model pretrained on electronic health records (EHR) and clinical notes, such as those in the MIMIC-III database, designed to understand and analyze medical text.

ClinicalBERT adapts the standard BERT architecture (12 transformer layers, 768-hidden units) to understand context-heavy clinical jargon

Attention!

Attention layers, specifically self-attention, refine static token embeddings into contextual representations by calculating weighted relevance between all input tokens.

They transform input embedding vectors into Query, Key, and Value matrices to compute attention scores, allowing each word to interact with others and capture contextual relationships.

MultiModal ClinicalBERT + Attention Classifier

Model's Architecture

Text branch:

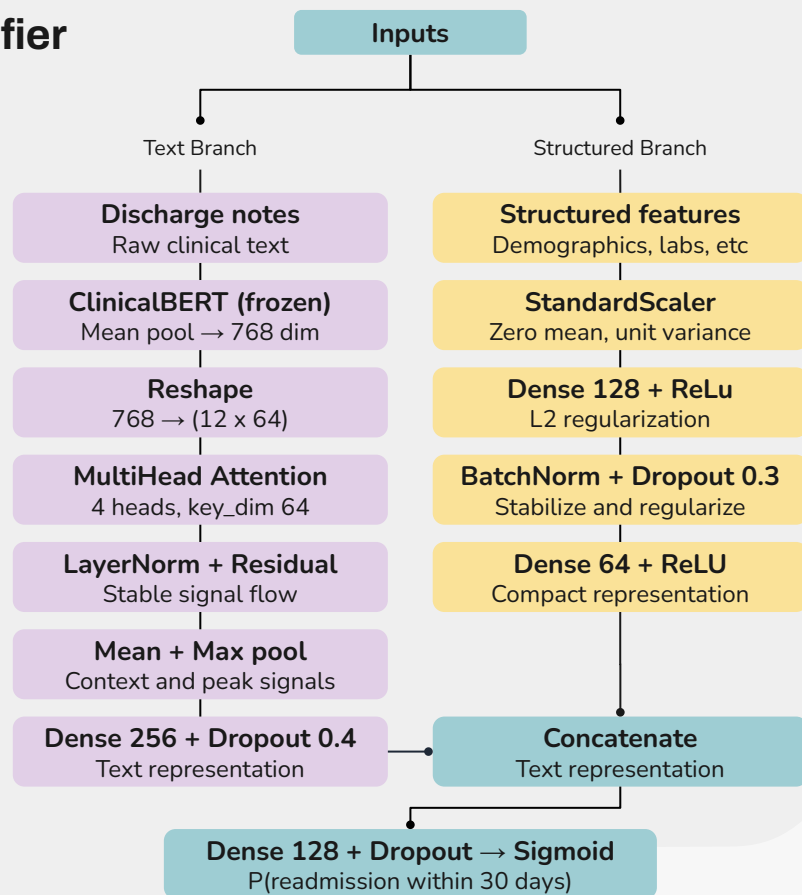
1. Turn flat embedding into a sequence of 12 "pieces" of 64 dimensions each so attention has something to operate over
2. Each head learns to weight relationships between different regions of the embedding differently
3. Standard transformer-style stabilization, lets the model fall back to the raw embedding if attention isn't helping
4. Dual pooling captures both overall context (mean) and peak signals (max). Balancing both scopes.
5. Regularization prior to concatenate

Structured branch:

1. Normalization
2. 128-dim ReLu activation + regularization
3. Normalization + regularization
4. 64-dim ReLu activation

Output:

1. Concatenate
2. 128-dim ReLu + Normalization + Regularization
3. Sigmoid activation output



AUC: 0.703 | Precision: 0.35 | Recall: 0.63 | F1: 0.45



04

Results & Comparison

+ Comparison Table

Metrics	RoBERTA	SBERT	ClinicalBERT
Test AUC	0.69	0.69	0.71
PR AUC	0.32	0.42	0.44
Recall	0.52	0.53	0.70
F1	0.44	0.43	0.46
Accuracy	0.62	0.65	0.66

Benchmarks

Other Research*
~ 0.59 - 0.76 AUC

Coin flip
50% accuracy roughly

*<https://pmc.ncbi.nlm.nih.gov/articles/PMC11271049/>



+ What ClinicalBERT actually catches

Confusion matrix

	Pred: Not readmit	Pred: Readmit
Actual: Not readmit	TN 25,725	FP 16,614
Actual: Readmit	FN 3,669	TP 8,561

CLINICAL READING

Catches 70%

Of true readmissions

Misses 3,669

True readmissions → patient deteriorates w/o intervention

Overflags 16,614

Home health visits and post-acute care coordination

Asymmetric trade

Accept more false alarms to catch more real cases



+ Conclusions & Future Work

KEY TAKEAWAYS

1

Domain matters more than scale

ClinicalBERT beats SBERT despite SBERT being a well-tuned general-purpose encoder. The domain prior is worth more than careful architecture tuning.

2

Attention + chunking lifted both encoders

SBERT's F1 climbed from 0.26 (Model 1, no chunking) to 0.43 (Model 6, chunking + attention); ClinicalBERT V1's vanilla mean pool reached 0.68 AUC, while V4's chunked encoding pushed it to 0.71.

3

Model is decision support, not a verdict

At our 0.5 threshold, the model catches 70% of true readmissions but overflags 16,614 patients who won't return. A discharge nurse calling 2 patients to catch 1 real readmission is cheap; missing that 1 patient is expensive.

FUTURE WORK

Fine-tune ClinicalBERT

Unfreeze top layers and fine-tune on the readmission task

Add temporal features

Time-series of vitals/labs over stay, not just summary stats

Threshold disclosure

Platt scaling so probabilities reflect real risk

External validation

Test on second institution's data to check generalization



Thanks

Any Questions?

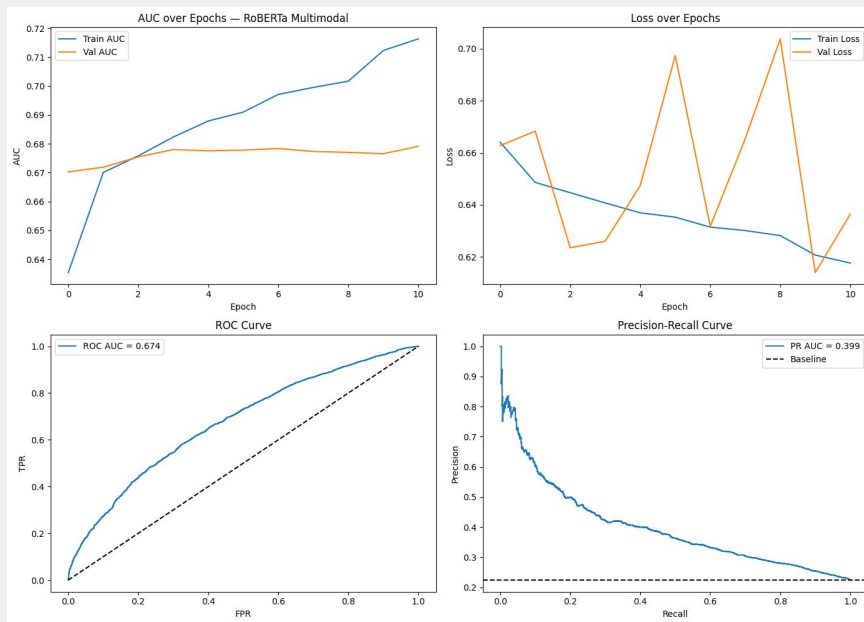




05

Appendix

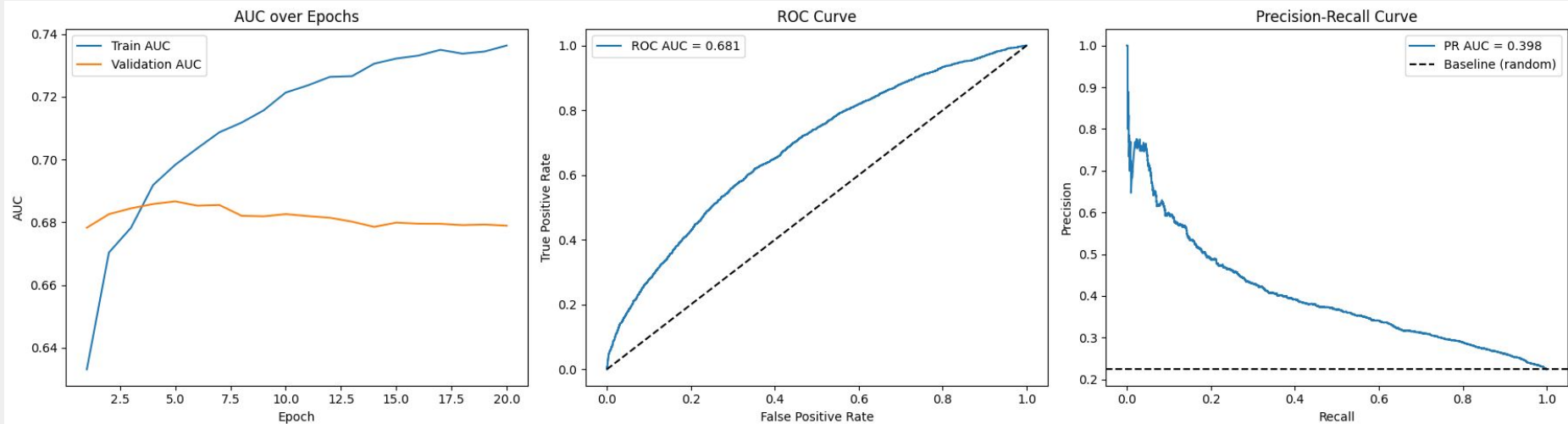
RoBERTa - Best Model



AUC: 0.69 | Precision: 0.37 | Recall: 0.52 | F1: 0.44



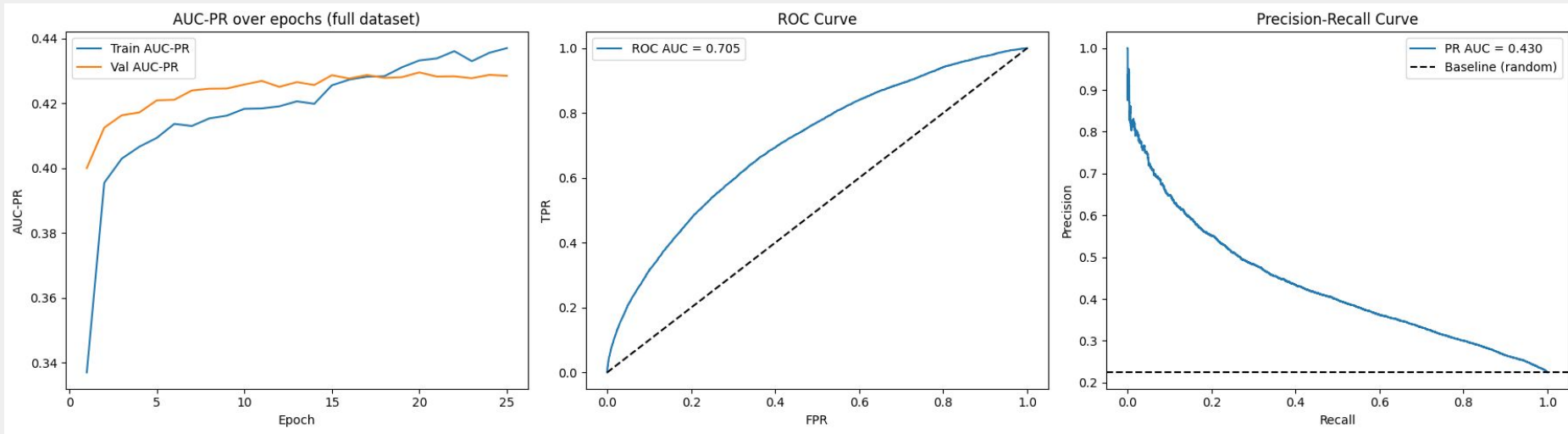
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